



PRODUCT NAME

Advanced Plastic Cleaner Formula

Description

A cost-effective, non-abrasive, and efficient liquid cleaner specially formulated for plastic surfaces. It provides excellent cleaning without compromising the integrity of the material.

Technical Specifications

pH Level:	7
Viscosity:	10-30 cP (low viscosity)
Shelf Life:	12-18 months
Batch Size:	10-1000 liters
Processing Time:	2-3 hours
Temperature:	Room temperature
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing tank, Stirrer, pH meter, Viscosity meter, Filtration system
Certification:	ISO 9001:2015, GMP

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	70.0%	Solvent
2	Sodium Lauryl Sulfate	Sodium Lauryl Sulfate	10.0%	Surfactant
3	Cocamidopropyl Betaine	Cocamidopropyl Betaine	8.0%	Surfactant
4	Sodium Citrate	Sodium Citrate	5.0%	pH Buffer
5	Glycerin	Glycerin	3.0%	Humectant
6	Citric Acid	Citric Acid	1.0%	pH Adjuster
7	Preservative Blend	Phenoxyethanol, Ethylhexylglycerin	1.0%	Preservative
8	Sodium Chloride	Sodium Chloride	2.0%	Viscosity Modifier
Total			100%	

Manufacturing Process

Phase 1: Preparation

1. Ensure all equipment is clean and sanitized.
2. Measure and prepare all ingredients according to the formulation.

Phase 2: Mixing

1. Add Water to the mixing tank.
2. Slowly mix in Sodium Lauryl Sulfate and Cocamidopropyl Betaine until well combined.
3. Add Sodium Citrate and ensure complete dissolution.

Phase 3: Adjustment

1. Incorporate Glycerin into the mixture.
2. Adjust pH to 7 with Citric Acid, monitoring pH continuously.
3. Mix in Preservative Blend thoroughly.

Phase 4: Finishing

1. Add Sodium Chloride to achieve the desired viscosity.
2. Final check for pH, viscosity, and homogeneity.
3. Filter the solution to remove any particulates.

Required Equipment

Mixing tank, Stirrer, pH meter, Viscosity meter, Filtration system

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: ISO 9001:2015, GMP

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10-1000 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply the cleaner directly onto the plastic surface with a cloth or spray bottle. Wipe with a clean, lint-free cloth. Repeat if necessary for stubborn stains. Do not rinse.